

DIAZEPAM

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Analytical Profile - Diazepam

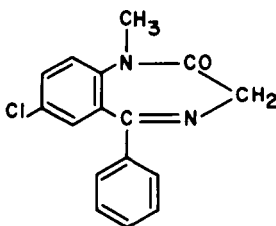
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1. Description

1.1 Name, Formula, Molecular Weight

Diazepam is 7-chloro-1,3-dihydro-1-methyl-5-phenyl-2H-1,4-benzodiazepin-2-one.



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$C_{16}H_{13}ClN_2O$

Mol. Wt. 284.75

1.2 Appearance, Color, Odor

Off-white to yellow, practically odorless, crystalline powder.

2. Physical Properties

2.1 Infrared Spectrum

The infrared spectrum of reference standard diazepam is presented in Figure 1¹. The spectrum was measured in a KBr pellet which contained 1 mg/400 mg KBr.

The following bands (cm^{-1}) have been assigned for Figure 1².

- Characteristic for NH:3390
- Characteristic for -C=O:1680
- Characteristic for aromatic groups:1560, 1480

2.2 Nuclear Magnetic Resonance Spectrum

The NMR spectrum shown in Figure 2 was obtained by dissolving 47 mg of reference standard diazepam in 0.5 ml of $CDCl_3$ containing tetramethylsilane as internal

Figure 1

Infrared Spectrum of Diazepam

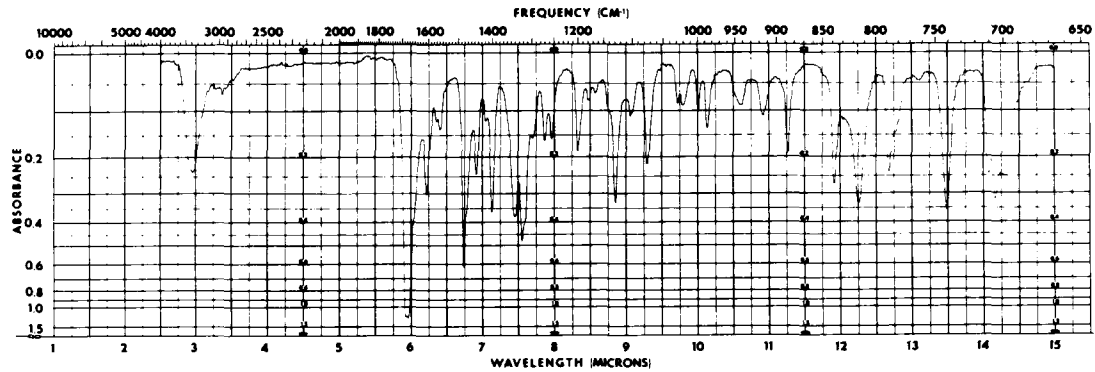
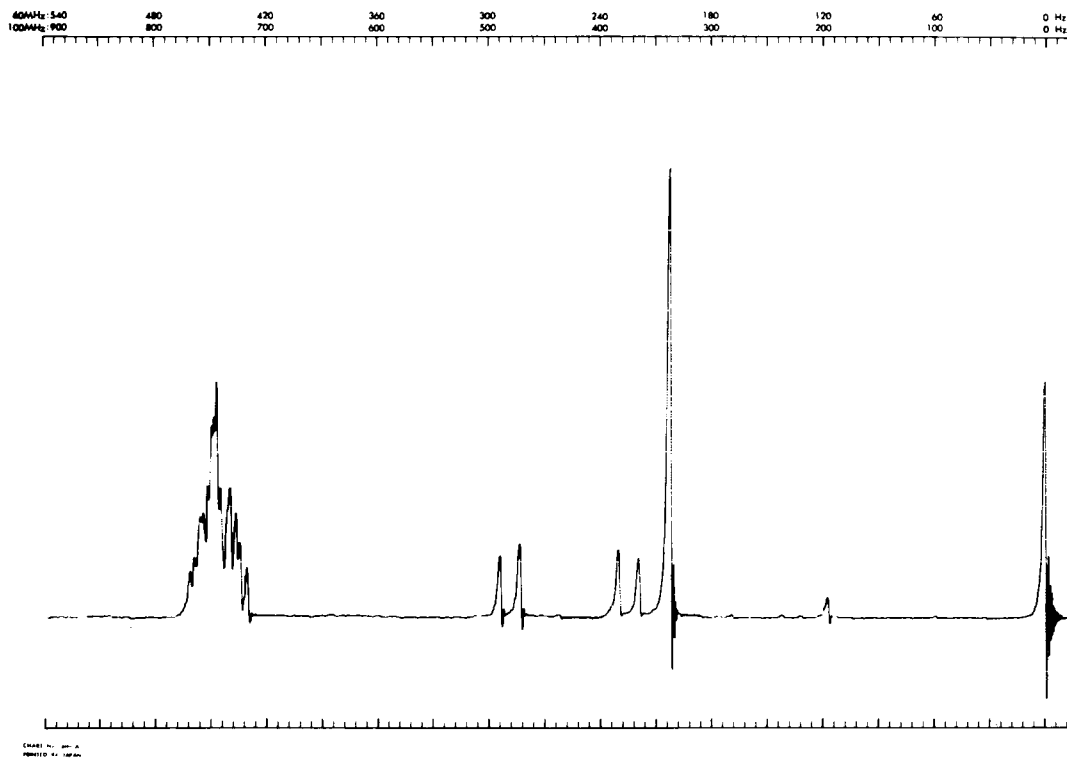


Figure 2

NMR Spectrum of Diazepam



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reference. The spectral assignments are shown in Table I³. Nuhn and Bley⁴ reported that at room temperature in C₆D₆ both methylene protons exhibit an AB spectrum. With increasing temperature the doublet was converted to a single line indicating rapid inversion of the ring. This observation was confirmed by Linscheid and Lehn⁵.

TABLE I

Diazepam

Protons at	Chemical Shift τ (ppm)	Type (J in Hz)
C ₁ methyl	6.62	s
C ₃ (a)	5.19	d(11)
C ₃ (b)	6.25	d(11)
C ₆ ,C ₈ ,C ₉ ,Ph	2.55	m

s = singlet; d = doublet; m = multiplet

2.3 Ultraviolet Spectrum

Diazepam when scanned between 420 and 210 nm in acidified 3A alcohol exhibits 3 maxima as shown in Figure 3. These were located at 242 ± 2 nm ($a = 100$), 285 ± 2 nm ($a = 43.7$) and 368 ± 2 nm ($a = 14.5$). Minima were observed at 221 ± 2 nm, 266 ± 2 nm and 334 ± 2 nm⁶.

2.4 Mass Spectra

The mass spectrum shown in Figure 4 was obtained using a CEC 21-110 mass spectrometer with an ionizing energy of 70 eV and a temperature of 190°C. Table II lists the elemental compositions for the most diagnostic ions as determined by high resolution mass spectrometry⁷. The molecular ion for diazepam was observed at m/e 284. The ions at m/e 256 and m/e 255 correspond to a loss of CH₂N and HCO respectively with the loss of chlorine shown by the ion m/e 249. Other ions in Table II can be ascribed to losses of Cl, and parts of the seven membered ring.

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Figure 3
Ultraviolet Spectrum of Diazepam

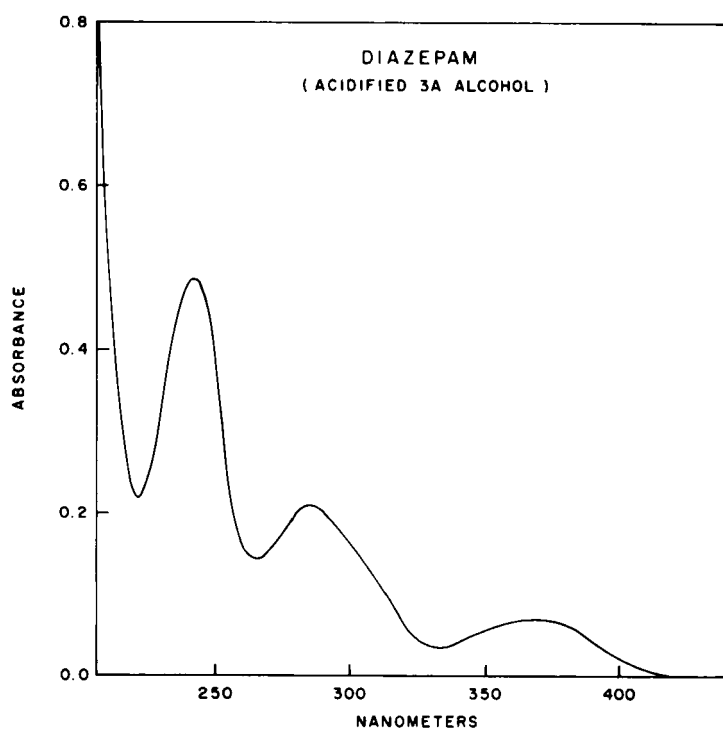
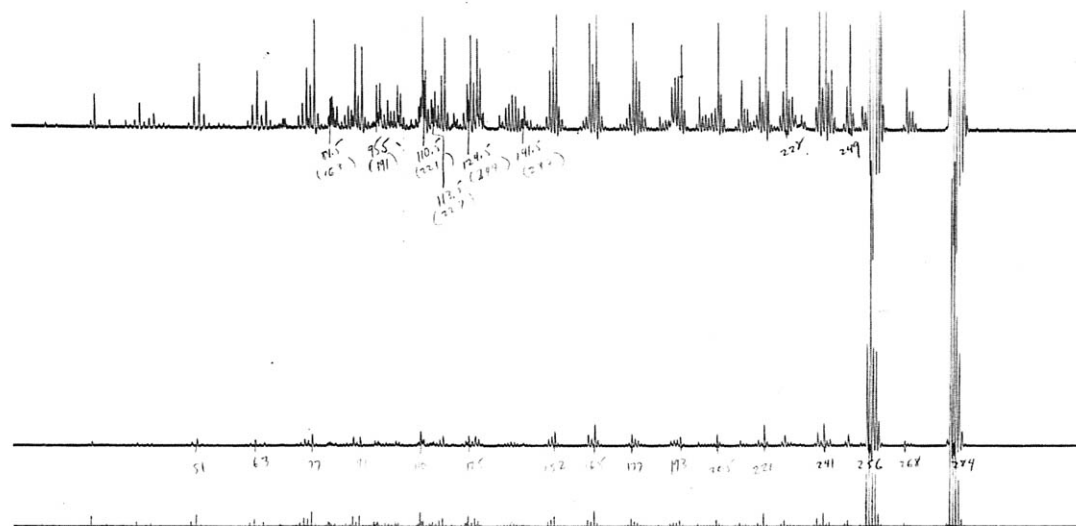


Figure 4
Mass Spectrum of Diazepam



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TABLE II

High Resolution Mass Spectrum of Diazepam^(a)

Found Mass	Calcd. Mass	<u>C</u>	<u>H</u>	<u>N</u>	<u>O</u>	<u>Cl</u> ₃₅
284.0681	284.0759	16	13	2	1	1
283.0630	283.0681	16	12	2	1	1
256.0521	256.0572	15	11	1	1	1
255.0680	255.0687	15	12	2	0	1
249.1009	249.1071	16	13	2	1	0
241.0520	241.0531	14	10	2	0	1
239.0374	239.0375	14	8	2	0	1
228.0582	228.0578	14	11	1	0	1
221.0817	221.0884	15	11	1	1	0
219.0917	219.0920	15	11	2	0	0
213.0353	213.0344	13	8	1	0	1
205.0758	205.0764	14	9	2	0	0
199.0300	199.0313	13	8	0	0	1
186.0236	186.0235	12	7	0	0	1

(a) Only peaks discussed are included in this table.

2.5 Optical Rotation

Diazepam exhibits no optical activity.

2.6 Melting Range

The melting range reported in NF XIII is 131-135°C.

2.7 Differential Scanning Calorimetry

The DSC spectrum of diazepam is shown in Figure 5. The endotherm observed at 129°C corresponds to the melt with a ΔH_f of 5.9 kcal/mole⁸. The decomposition temperature is 180°C.

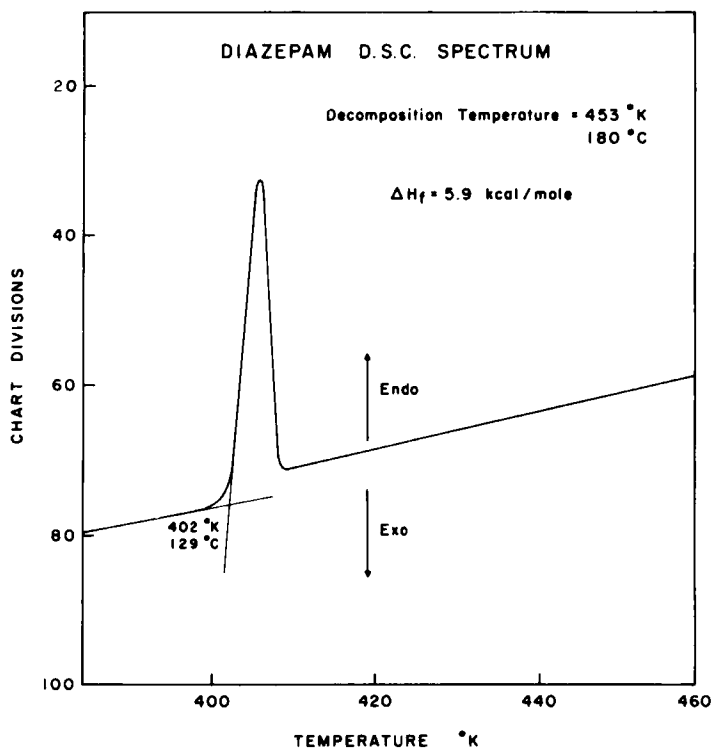
2.8 Thermogravimetric Analysis

A thermal gravimetric analysis performed on diazepam exhibited no loss of weight when heated to 105°C⁸.

2.9 Solubility

Approximate solubility data obtained at room temperature are given in the following table.

Figure 5
Diazepam D.S.C. Spectrum



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<u>Solvent</u>	<u>Solubility mg/ml</u>
Water	.05
Petroleum Ether (30°-60°)	.9
Propylene Glycol	17
Ether	18
Isopropanol	20
3A Alcohol	32
95% Ethanol	41
Methanol	49
Acetone	125
Benzene	220
Dimethylacetamide	296
Chloroform	>500

2.10 Crystal Properties

The x-ray powder diffraction pattern of diazepam is presented in Table III⁹.

Instrument Conditions

General Electric	Model XRD-6 Spectrogoniometer
Generator	50 KV, 12-1/2 MA
Tube target	Copper
Optics	0.2° Detector Slit
	MR soller slit
	3° Beam slit
	0.0007" Ni Filter
	4° take off angle
Goniometer	Scan at 0.4° 20 per minute
Detector	Amplifier - 16 coarse, 8.7 fine (gain)
	Sealed proportional counter tube and DC voltage at plateau
	Pulse height selector E _L - 5 volts; Eu - out
	Rate meter T.C. 4 2000 C/S full scale
Recorder	Chart speed - 1 inch per 5 minutes

Samples prepared by grinding at room temperature.

TABLE III

Diazepam

2θ	$d(A)^*$	I/I_0^{**}
9.44°	9.37	15%
11.00	8.04	7
13.20	6.71	3
13.60	6.51	18
14.60	6.07	3
17.20	5.16	5
17.52	5.06	10
18.88	4.70	100
20.96	4.24	3
21.56	4.12	6
22.04	4.03	15
22.80	3.90	65
23.80	3.74	27
24.44	3.64	25
26.12	3.41	4
26.68	3.34	13
26.88	3.32	10
27.48	3.25	11
28.32	3.15	2
28.88	3.09	2
29.44	3.03	14
29.68	3.01	16

$$*d = \frac{n}{2} \frac{\lambda}{\sin \theta} \text{ (interplanar distance)}$$

** I/I_0 = relative intensity (based on highest intensity of 1.00)

2.11 Dissociation Constant

The pKa for diazepam has been determined spectrophotometrically to be 3.4¹⁰.

2.12 Distribution Coefficient

The distribution coefficient of diazepam between 1-octanol and pH 7.2 phosphate buffer is 382 at room temperature where $D = C_{\text{octanol}}/C_{\text{buffer}}$ ¹⁰.

3. Synthesis

Diazepam may be prepared by the reaction scheme shown in Figure 6 with the reaction of 2-methylamino-5-chloro-benzophenone and ethyl glycinate to form diazepam¹¹. A complete review of the chemistry of benzodiazepines describes other synthetic routes¹².

4. Stability-Degradation

The acid hydrolysis products for both diazepam and its major metabolite are shown in Figure 7. The acid hydrolysis of chlordiazepoxide is also included since it is the same as for the major diazepam metabolite¹³.

5. Drug Metabolic Products and Pharmacokinetics

The major metabolites of diazepam in humans are shown in Figure 8¹⁴. The major metabolic pathways were shown to consist of demethylation at the nitrogen in position 1, addition of a hydroxyl group at carbon 3, and conjugation of the respective derivatives. The primary metabolite in blood is the N-desmethyl diazepam and in urine the oxazepam-glucuronide. Analytical procedures for the metabolites have been published using ultraviolet spectrometry¹⁴, thin-layer chromatography¹⁴ and gas chromatography^{14,15}.

6. Methods of Analysis

6.1 Elemental Analysis

<u>Element</u>	<u>% Theory</u>	<u>Reported</u>
C	67.49	67.33
H	4.60	4.63

6.2 Phase Solubility

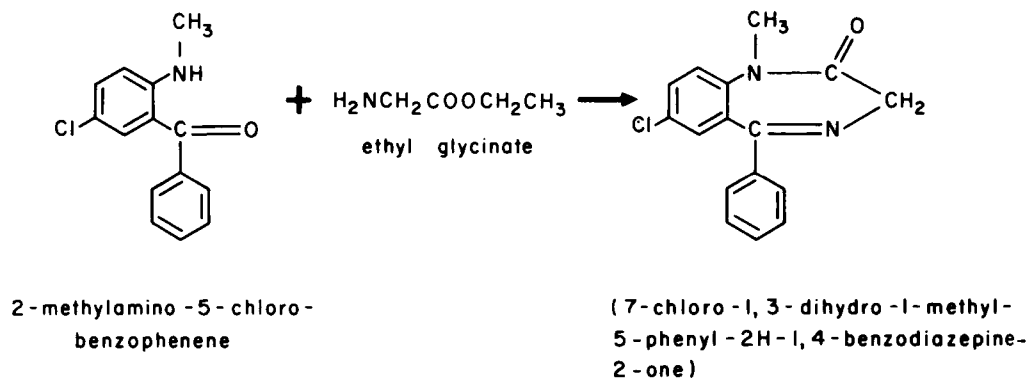
Phase solubility analysis is carried out using isopropanol:hexane (1:1) as the solvent. A typical example is shown in Figure 9 which also lists the conditions under which the analysis was carried out¹⁶.

6.3 Chromatographic Analysis

6.3.1 Thin-Layer Chromatographic Analysis

The following TLC system is useful as an identity test and evaluation of diazepam substance. Using silica gel G plates and a mixed solvent system of

Figure 6
Synthesis of Diazepam



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Figure 7
Hydrolysis of Diazepam

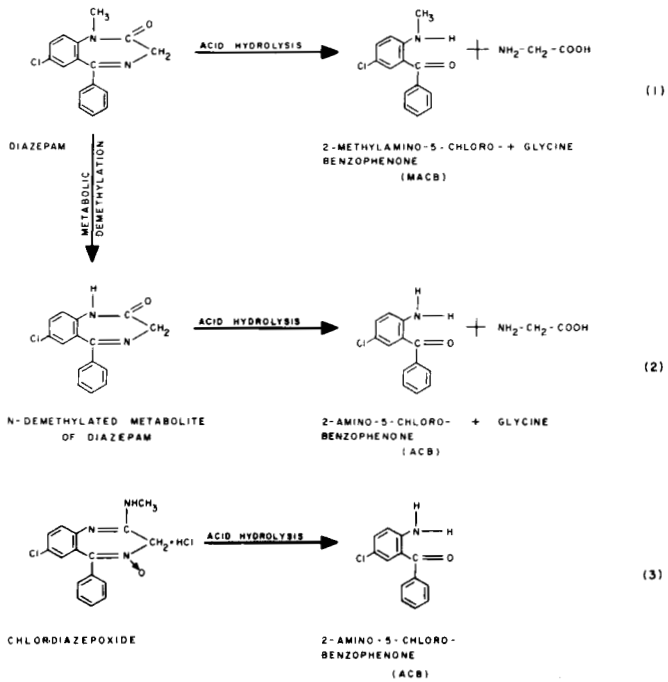


Figure 8
Metabolism of Diazepam

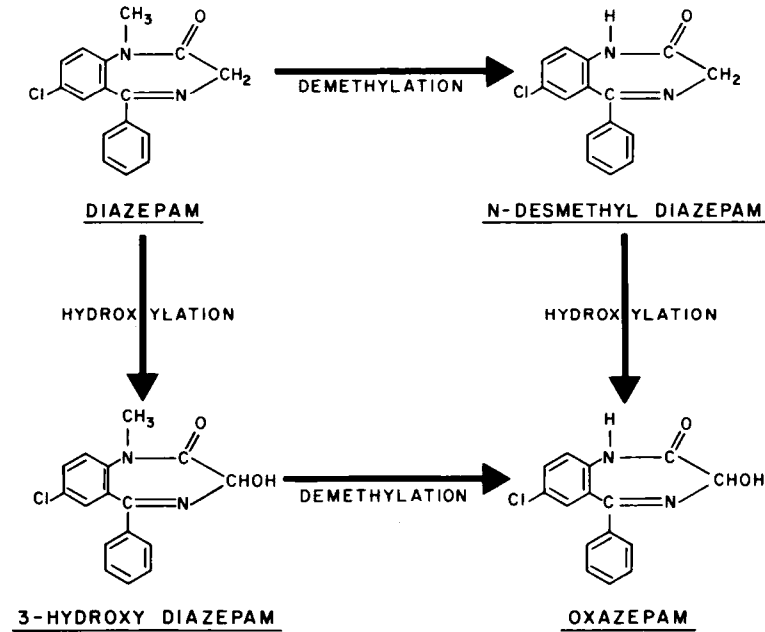
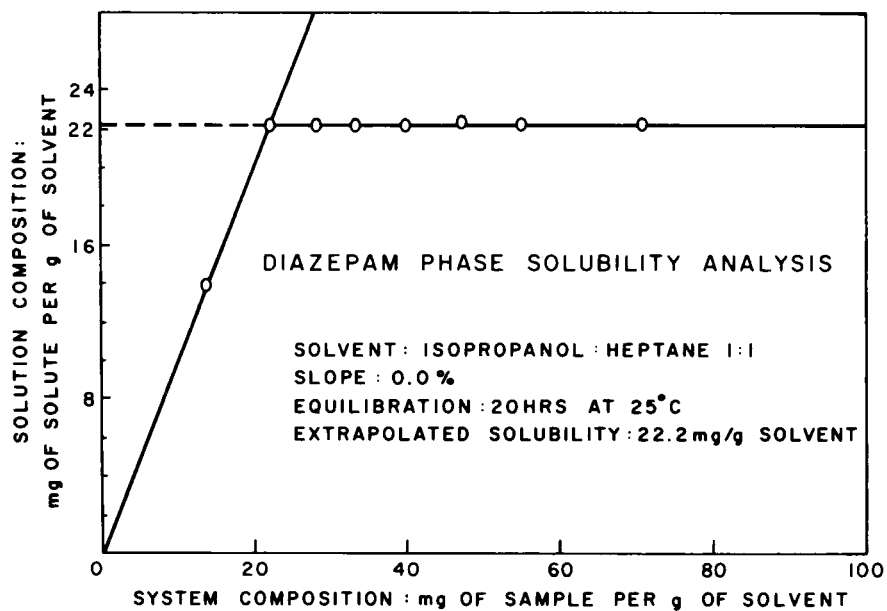


Figure 9

Phase Solubility Analysis of Diazepam



n-heptane:ethyl acetate (1:1 v/v), the sample containing 0.5 mg of diazepam substance in acetone is spotted and subjected to ascending chromatography. After development for at least 10 cm, the plate is air-dried and sprayed with potassium iodoplatinate solution. Diazepam appears as a purple spot with an approximate R_f of 0.3-0.4¹⁷. The second system is useful for identification of diazepam and its metabolites in extracts of biological materials^{14,15}. The method uses two dimensional development of Brinkmann [F254] precoated silica gel plates using chloroform:heptane:ethanol (10:10:1 v/v) for the first dimension and chloroform:acetone (90:10 v/v) for the second dimension. The R_f ranges for each compound with the respective systems are tabulated below. The spotting solvent is acetone:hexane (20:80 v/v), the final extract.

TABLE IV

Compound	<u>R_f</u>			<u>R_f</u>	
	System I			System II	
	Chloroform:Heptane:Ethanol			Chloroform:Acetone	
	10	10	1	90	10
Diazepam	0.38-0.43			0.39-0.41	
N-Desmethyl-diazepam	0.20-0.23			0.16-0.21	
3-Hydroxy-diazepam	0.29-0.30			0.33-0.33	
Oxazepam	0.09-0.11			0.08	

6.32 Column Chromatographic Analysis

The analytical scale liquid-solid chromatographic separation of diazepam and its metabolites has been reported by Scott and Bommer in their study of the separation of benzodiazepines from each other and from biological media¹⁸. The liquid-solid chromatography was carried out using Durapak "OPN" (36-75 μ particle diameter) 100 cm column with a 1 mm inside diameter and hexane:isopropanol (95:5 v/v) as solvent. The flow rate was 1.0 ml/min using an air driven pump. The detector was an ultraviolet monitor set at 254 nm. The

cell volume was 8 μ l. Complete separation of diazepam, n-desmethyl diazepam, 3-hydroxy diazepam and oxazepam is reported at 2 μ g sensitivity per compound.

6.33 Vapor Phase Chromatography

The acid hydrolysis of blood extracts containing diazepam and its metabolite (Figure 7) has been used by deSilva¹³ to prepare the respective benzophenones as volatile derivatives for gas chromatography. The column used was a 2 foot, 1/4", 2% Carbowax 20 M on silanized Gas Chrom P at 190° using a tritium electron capture detector. The disadvantages of this approach i.e. the benzophenones obtained in acid hydrolysis are not specific for a single benzodiazepine have been eliminated in a recent paper by deSilva¹⁵ where the benzodiazepines have been chromatographed directly.

The direct method uses a 4 foot, 4 mm, column at 230°C packed with 3% OV-17 on 60/80 mesh Gas Chrom Q. The detector was a Ni⁶³ electron capture detector operated at 310°. The use of stable high temperature phases and the high temperature Ni⁶³ detector has enabled the quantitation of intact diazepam and its metabolites in blood and urine extracts at the following sensitivities.

<u>Compound</u>	<u>Sensitivity, nanograms</u>
Diazepam	1.0
N-desmethyl diazepam	2.0
Oxazepam	1.0

6.4 Direct Spectrophotometric Analysis

Direct spectrophotometric analysis of diazepam is applicable provided significant quantities of the hydrolytic contaminants are not present. For material not containing the interfering species the reported maxima at 368 + 2 nm in 0.1N alcoholic sulfuric acid with an absorptivity value of 14.5 may be used for quantitative measurement. The Technicon Autoanalyzer system for dosage form assays of diazepam is based on the direct spectrophotometric assay.

6.6 Polarographic Assay

A single reduction wave for diazepam has been observed by several investigators in aqueous system^{19,20,21}. The single wave has been attributed to the reduction of the -C=N- moiety and the diffusion current is proportional to concentration in the range of 2×10^{-4} to 7×10^{-4} M.

6.7 Non-Aqueous Titration

Diazepam may be titrated in acetic anhydride using HClO_4 in glacial acetic acid and nile blue hydrochloride indicator. Each ml of 0.1N HClO_4 is equivalent to 28.48 mg of diazepam¹⁷.

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